Nyaya Saathi: AI Lawyer (Kuku Assignment, Problem 2)

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Problem and scope

I picked Problem 2 because its hardest part is a design question: how to stop an AI from inventing law. I went deep on three everyday money disputes instead of covering everything: **a security deposit not returned**, **a refund not received or a defective product**, and **unpaid salary**, plus **explaining a notice** someone received. For each, it builds the case, says where the person stands, drafts a legal notice (.docx) and lets them practise arguing with the other side. **Not covered, on purpose:** criminal and police matters, domestic violence, divorce, property title, cyber fraud and cases already in court. For these it gives helplines (112, 181, 1930) or free legal aid (NALSA 15100) instead of guessing.

Assumptions: the user is an adult in India writing or speaking English, Hindi or Hinglish. The goal is action (a notice, a complaint), not a verdict. A wrong law is worse than "I'm not sure". Notices are drafted in English. The law is as of November 2026, including the BNS (July 2024) and the Labour Codes (November 2025).

How it works

Every message, typed or spoken (Whisper), goes through seven steps:

1. **Safety rules (code):** keywords in three languages catch violence, self-harm, fraud, police and court. Danger stops the chat and shows helplines; police or court adds a "see a lawyer" banner.
2. **Triage (AI):** finds the problem type and language. Out-of-scope problems get an honest refusal.
3. **Case builder (AI + code):** the AI extracts amount, dates, parties and proof; code lists what's missing; the AI asks for it, at most 2 questions per round and 2 rounds.
4. **Calculations (code):** days passed, the 2- and 3-year limits, the consumer forum by amount, the filing fee.
5. **Advice (AI):** may use only 19 verified knowledge-base entries, and must tag each point with the entry it relies on.
6. **Validator (code):** deletes court cases, replaced laws (IPC, Payment of Wages Act), unknown sections, phone numbers and websites, and tells the user what it removed.

7. **Output (code):** a fixed layout with sources, caveats, a confidence level set by code, and when to see a lawyer. In the notice, the AI writes only the facts; the legal basis comes from a template.

The split: the AI understands and explains; code holds everything that must be exact.

Scale, cost and humans in the loop

A conversation takes about 4.5 model calls and 6,000 tokens: free on Groq, roughly ₹0.15–0.20 at paid prices (approximate). Every provider uses the same OpenAI-compatible API, so swapping the model is one line in `config.yaml`; I ran the tests on two models. A new legal area needs one knowledge-base file, one case form and a few tests. A lawyer should sign off each knowledge-base entry, and a weekly sample of conversations plus the validator's log should go to a reviewer.

How I know it's good, and where it fails

17 scripted conversations (everyday cases in three languages, notices, red flags, IPC and case-law traps, out of scope) run through a script that saves the raw outputs and checks problem type, language, escalation, the lawyer flag and citations. 16 unit tests cover the safety code. On `gpt-oss-20b` with the final code, all 13 cases that finished passed; 4 stopped at Groq's daily limit. An earlier `gpt-oss-120b` run passed 17/17 before the last fixes.

Reading the raw outputs caught what the checks missed. It skipped follow-up questions (fixed in code). It claimed "Maharashtra has not adopted the Model Tenancy Act", which the validator can't catch because it checks citations, not every sentence (**still open**). It guessed unpaid months into a notice (prompt fixed; the user must still read the draft). The smaller model answered an English user in Hindi (code now detects this and retries). In real use I would track notices sent, complaints filed and "this was wrong" reports.

AI tools used

Claude (Claude Code) to plan, write code and check legal facts; Groq's `gpt-oss-120b`, `gpt-oss-20b` and Whisper inside the product. The decisions on scope, what stays in code and when to refuse were mine. Claude got things wrong: it picked a retired default model, and its validator blocked the correct e-jagriti.gov.in link because of an unusual hyphen. Both were caught only by running the real tests.

Links: Prototype · GitHub code · Test cases and raw outputs · Walkthrough video